

Solutions to Mock JEE MAIN - 20 | JEE 2022

PHYSICS

SECTION-1

- 1.(B) Using energy conservation

$$-\frac{GM_e m}{2R_e} + \frac{1}{2}m \left(\sqrt{\frac{5GM_e}{2R_e}} \right)^2 = \frac{1}{2}mv^2$$

$$v = \sqrt{\frac{3GM_e}{2R_e}}$$

- 2.(C) Let radius of circle is R .

$$\text{Field due to square at the centre is } B_1 = 4 \times \frac{\mu_0 i_1}{4\pi R} (\cos 45^\circ + \cos 45^\circ) = \frac{\sqrt{2}\mu_0 i_1}{\pi R}$$

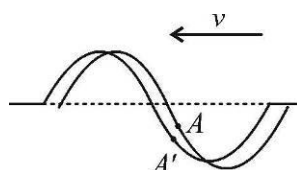
$$\text{Field due to ring at the centre is } B_2 = \frac{\mu_0 i_2}{2R}$$

They must cancel each other in order to make net magnetic field at the centre zero

So, $B_1 = B_2$

$$\frac{\sqrt{2}\mu_0 i_1}{\pi R} = \frac{\mu_0 i_2}{2R} \Rightarrow i_1 / i_2 = \frac{\pi}{2\sqrt{2}}$$

- 3.(D) Referring to the diagram right wave is initial position and left wave is final position. It is clearly visible that the particle is moving towards extreme position and in SHM acceleration is always towards mean position.



- 4.(A) Because stopping potential of A is greater than B so frequency of A must also be greater than B . Saturation current of B is greater than A so Intensity of B should be greater than A .

- 5.(D) Let C is capacitance of capacitor its value will be $C = \frac{\epsilon_0 A}{d}$ (i)

After filling the wax and doubling plate separation $2C = \frac{K\epsilon_0 a}{2d}$ (ii)

On solving $K = 4$

- 6.(C) Area of rhombus will be $= \ell^2 \sin \theta$

So flux through the rhombus will be $\phi = B_0 \ell^2 \sin \theta$

So Emf induced in the shape will be $E = -\frac{d\phi}{dt} = -B_0 \ell^2 \cos \theta \left(\frac{d\theta}{dt} \right)$

To calculate the value of $\frac{d\theta}{dt}$ we can say distance between A & C is changing with $2v$

$$\text{So } 2v = \frac{d}{dt} (2\ell \cos \theta / 2)$$

$$2v = \frac{2\ell}{2} \left(-\sin \left(\frac{\theta}{2} \right) \right) \frac{d\theta}{dt}$$

$$\text{So } \frac{d\theta}{dt} = -\frac{2v}{\ell \sin \theta / 2}$$

Replacing the value of $\frac{d\theta}{dt}$

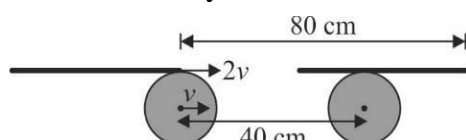
$$\text{We get } E = \frac{2B_0 \ell v \cos \theta}{\sin \theta / 2}$$

7.(D) According to the question

$$E_1 + E_2 \propto 72 \quad \& \quad E_1 - E_2 \propto 54$$

$$\text{So } \frac{E_1 + E_2}{E_1 - E_2} = \frac{72}{54} = \frac{4}{3} \Rightarrow \frac{E_1}{E_2} = \frac{7}{1}$$

$$8.(C) \int v dt = 40 \text{ cm} \Rightarrow \int 2v dt = 80 \text{ cm}$$



$$\therefore \ell = 80 \text{ cm} - 40 \text{ cm} = 40 \text{ cm}$$

9.(A) Moment of inertia of disc will be $I = \frac{MR^2}{2}$

Because disc is converted into sphere so radius of sphere will be R'

$$\text{So, } \frac{4}{3} \pi R'^3 = \pi R^2 \times \frac{R}{6}$$

$$R' = \frac{R}{2}$$

$$\text{So, moment of inertia of sphere will be } I_s = \frac{2}{5} M \left(\frac{R}{2} \right)^2 = \frac{MR^2}{10}$$

$$\text{So, } I_s = \frac{I}{5}$$

10.(D) Total energy of system

$$v = \frac{kq \times -2q}{a} \times 2 + \frac{kqq}{2a}$$

$$= -\frac{4kq^2}{a} + \frac{kq^2}{2a} = -\frac{4kq^2}{a} + \frac{kq^2}{2a} = -\frac{7kq^2}{2a}$$

$$11.(D) KE = \frac{3}{2} nRT = \frac{3}{2} PV \Rightarrow P = \frac{2}{3} \frac{KE}{V} = \frac{2}{3} E$$

$$12.(B) \text{ We know } t_{\text{mean}} = \frac{1}{\lambda} = t_0 \quad \text{and} \quad t_{1/2} = \frac{\ell n 2}{\lambda} = t_0 \ell n 2$$

Hence $\frac{1}{2}$ of the sample remains

$$13.(B) \sigma 2\pi r \cos \theta = Mg$$

$$\sigma 2\pi \frac{r}{2} \cos \theta = M'g$$

$$\therefore M' = \frac{M}{2}$$

14.(C) We know current leads by $\frac{\pi}{2}$ angle

$$\text{So, } i = i_0 \cos\left(\omega t + \frac{\pi}{2}\right) = -i_0 \sin \omega t$$

15.(A) We can say $T = k(S)^a (r)^b (\rho)^c \Rightarrow [T] = [MT^{-2}]^a [L]^b [ML^{-3}]^c$

$$\text{We can say } a + c = 0$$

$$b - 3c = 0$$

$$-2a = 1$$

$$\text{So } a = -\frac{1}{2}, c = +\frac{1}{2} \text{ \& } b = +\frac{3}{2}$$

$$\text{So } T = k(S)^{-1/2} (r)^{3/2} (\rho)^{1/2} \text{ or } T = k \sqrt{\frac{r^3 \rho}{S}}$$

$$16.(C) \frac{P_0 V_0}{RT_0} + \frac{P_0 V_0}{RT_0} = \frac{PV_0}{R2T_0} + \frac{PV_0}{RT_0}$$

$$P = \frac{4}{3} P_0$$

17.(D) The truth table is of AND gate

18.(B) Phase difference at a point at a distance x from the central maxima, $\Delta\phi = 2\pi\left(\frac{x}{\beta}\right)$

$$\text{Intensity at this point, } I = I_0 \cos^2\left(\frac{\Delta\phi}{2}\right) = I_0 \cos^2\left(\frac{\pi}{4}\right) = \frac{I_0}{2}$$

19.(A) We know $\vec{E} \times \vec{B} = \vec{C}$

Using right hand thumb rule $\vec{B}$ will be directed along $-z$ axis

$$\hat{i} \times (-\hat{k}) = \hat{j}$$

20.(C) Convectional current due to motion of electron will be $i = \frac{e}{T} = \frac{e\omega}{2\pi} = \frac{e}{2\pi} \frac{v}{r}$

$$\text{So magnetic field at the centre will be } B = \frac{\mu_0 i}{2r} = \frac{\mu_0}{4\pi} \frac{ev}{r^2}$$

SECTION-2

21.(5) $d\ell = \ell_0 \alpha dT$

$$\int_0^{\Delta\ell} d\ell = \ell_0 \alpha_0 \int_0^{25} TdT$$

$$\Delta\ell = \ell_0 \alpha_0 \times \frac{625}{2} = 10 \times 1.6 \times 10^{-6} \times \frac{625}{2} = 5 \times 1000 \times 10^{-6} = 5mm$$

22.(18) Using angular momentum conservation final angular velocity will be

$$3 \times 10 + 9 \times 6 = (3 + 9)\omega$$

$$\omega = 7 \text{ rad / sec}$$

$$\text{So loss of } KE = \frac{1}{2} \times 3 \times 10^2 + \frac{1}{2} \times 9 \times 6^2 - \frac{1}{2} \times 12 \times 7^2 = 18J$$

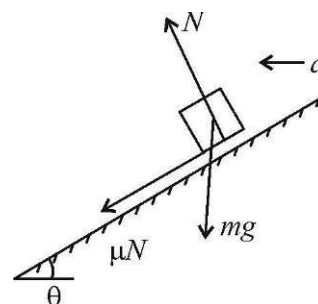
23.(20) From free body diagram in vertical direction we can say

$$N \cos \theta = mg + \mu N \sin \theta$$

$$N = \frac{mg}{\cos \theta - \mu \sin \theta}$$

$$\text{So, } a = \frac{g \sin \theta}{\cos \theta - \mu \sin \theta} + \frac{\mu g \cos \theta}{\cos \theta - \mu \sin \theta}$$

$$= g \left[\frac{\sin \theta + \mu \cos \theta}{\cos \theta - \mu \sin \theta} \right] = 20 \text{ m/sec}^2$$



24.(40) Let height of source of light is h_0 so for 1st magnification $|m_1| = \left| \frac{20}{h_0} \right|$

$$\text{And for 2nd magnification } |m_2| = \left| \frac{80}{h_0} \right|$$

$$\text{We know that } m_1 m_2 = 1 \quad \text{So, } h_0 = 40 \text{ cm}$$

25.(2.80) Excess pressure inside the water film $\Delta P = \frac{T}{d/2} = \frac{70 \times 10^{-3}}{0.05 \times 10^{-2} / 2} = 280 \text{ N/m}^2$

$$\text{So, Force} = \Delta P \times A = 280 \times 10^{-2} = 2.8 \text{ N}$$

26.(750) Case-1

$$f_{app} = f \frac{V}{V - V_s}; \quad 1000 = f \frac{350}{350 - 50}; \quad f = \frac{30}{35} \times 1000 \text{ Hz}; \quad f = \frac{6000}{7} \text{ Hz}$$

Case-2

$$f_{app} = f \frac{V}{V + V_s} = \frac{6000}{7} \cdot \frac{350}{400} = 750 \text{ Hz}$$

27.(37) For Brewster law $\tan i = \mu \Rightarrow i = 53^\circ$

$$\text{So angle of refraction } r = 90 - i = 37^\circ$$

28.(4) $B = \frac{\mu_0}{4\pi} \frac{2M}{r^3} = \frac{10^{-7} \times 2 \times 2.5}{(0.5)^3} = 40 \times 10^{-7} \text{ Tesla} = 4 \mu T$

29.(16) To make net electric field O electric field due to shell must be equal to electric field due to sphere

$$\text{so } \frac{KQ}{4R^2} = \frac{K \times q}{(3R/2)^2}; \quad \frac{KQ}{4R^2} = \frac{Kq}{9R^2} \times 4 \Rightarrow Q = 16$$

30.(20) Taking right direction positive.

$$\vec{v}_O = 10$$

$$\& \vec{v}_m = -5$$

$$\text{So } \vec{v}_{OM} = \vec{v}_O - \vec{v}_M = 15 \text{ m/sec}$$

$$\text{So } \vec{v}_{IM} = -15 \text{ m/sec}$$

$$\vec{v}_{IM} = \vec{v}_I - \vec{v}_M = -15$$

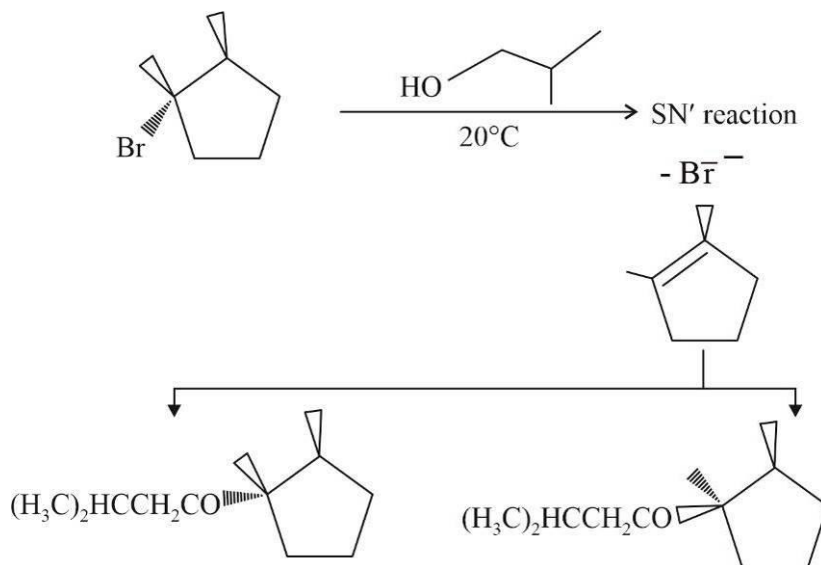
$$\vec{v}_I = -15 - 5 = -20 \text{ m/sec}$$

$$\text{So, speed of image is } 20 \text{ m/sec}$$

CHEMISTRY

SECTION-1

- 1.(C) Laundry soap contains fillers like sodium rosinate, sodium silicate, borax and sodium carbonate
 2.(C) $114 \rightarrow$ Un un quad ium, UUq
 3.(D) This is the general structure of penicillin drug which is used as antibiotic drugs.
 4.(B)



- 5.(C) peroxyacetyl nitrate is a secondary air pollutant
 6.(A) Solute and solvent both should obey Raoult's law
 7.(B) $3\text{Pt} + 4\text{HNO}_3 + 18\text{HCl} \rightarrow 3\text{H}_2\text{PtCl}_6 + 4\text{NO} + 8\text{H}_2\text{O}$

Total moles of e^- transferred = 12

So, e^- per mole of Pt = $\frac{12}{3} = 4$

- 8.(C) $\text{C}_2 \rightarrow (\sigma_{1s})^2 (\sigma_{1s}^*)^2 (\sigma_{2s})^2 (\sigma_{2s}^*)^2 (\pi_{2p_x})^2 (\pi_{2p_y})^2$

$$\text{B.O} = \frac{4-0}{2} = 2$$

$$\text{C}_2^- \rightarrow (\sigma_{1s})^2 (\sigma_{1s}^*)^2 (\sigma_{2s})^2 (\sigma_{2s}^*)^2 (\pi_{2p_x})^2 (\pi_{2p_y})^2 (\sigma_{2p_z})^1$$

$$\text{B.O} = \frac{5-0}{2} = 2.5$$

$$\text{C}_2^{2-} \rightarrow (\sigma_{1s})^2 (\sigma_{1s}^*)^2 (\sigma_{2s})^2 (\sigma_{2s}^*)^2 (\pi_{2p_x})^2 (\pi_{2p_y})^2 (\sigma_{2p_z})^2$$

$$\text{B.O} = \frac{6-0}{2} = 3$$

As bond order is increasing it implies bond strength is increasing.

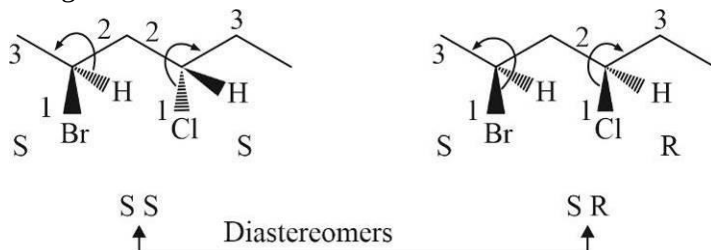
- 9.(D) At pH = 6.4
 As in case of H_2CO_3 ,
 pH = pKa it will be only when
 [weak acid] = [conjugate base].
 In case of $\text{H}_2\text{PO}_4^- | \text{HPO}_4^{2-}$

$$\text{pH} = \text{pK}_a + \log \frac{[\text{HPO}_4^{2-}]}{[\text{H}_2\text{PO}_4^-]}$$

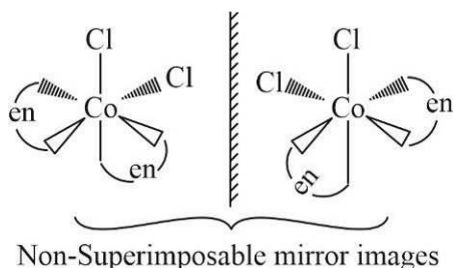
$$6.4 = 7.2 + \log \frac{[\text{HPO}_4^{2-}]}{[\text{H}_2\text{PO}_4^-]}$$

$$\log \frac{[\text{HPO}_4^{2-}]}{[\text{H}_2\text{PO}_4^-]} = -\text{ve} \quad ; \quad \frac{[\text{HPO}_4^{2-}]}{[\text{H}_2\text{PO}_4^-]} < 1 \Rightarrow [\text{HPO}_4^{2-}] < [\text{H}_2\text{PO}_4^-]$$

10.(D) Assign R & S nomenclature to chiral carbons



11.(C) cis-Co(en)₂Cl₂



12.(B) 1st order reaction is fastest at the starting of the reaction then it decrease also you can see the slope which is equal to rate.

13.(A) Order of electronegativity sp carbon $>$ sp^2 Carbon $>$ sp^3 carbon

14.(C) Reduction of alkyl cyanide using SnCl_2 and HCl is called as Stephen reduction reaction.

15.(D) Red phosphorus undergoes disproportionation on reaction with alkali to form $\text{H}_4\text{P}_2\text{O}_6$ and PH_3

$$16.(C) \quad \frac{x}{m} \propto P^{\frac{1}{n}} \Rightarrow \frac{x}{m} = k P^{\frac{1}{n}} \quad ; \quad \log\left(\frac{x}{m}\right) = \frac{1}{n} \log P + \log k$$

17.(A) $3d \rightarrow 4d \rightarrow 5d$ CFSE increases for the same ligands.

18.(A) N_2O & NO

19.(B) Partition

20.(D) ion flow only

SECTION-2

21.(84) From the principal of atomic conservation on carbon

$$3 \times \text{moles of } C_2H_5OH = 3 \times \text{moles of } HC_2H_3O_2$$

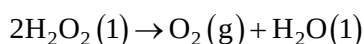
$$\text{moles of } HC_2H_3O_2 = \frac{5}{46} \text{ (Theoretical)}$$

$$\text{moles of } HC_2H_3O_2 \text{ (Experimental)} = \frac{5.5}{60}$$

$$\% \text{ yield} = \frac{5.5}{60 \times 5} \times 46 \times 100 = 84\%$$

$$22.(3) \quad X = \frac{1}{8} \times 8 = 1 ; \quad Y = 6 \times \frac{1}{2} = 3 ; \quad X_a Y_b = XY_3 \Rightarrow \frac{b}{a} = \frac{3}{1} = 3$$

23.(3) 10 volume solutions of H_2O_2 means that 1L of this H_2O_2 solution will give 10 L of oxygen at STP



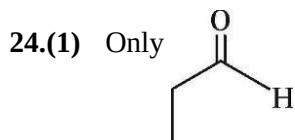
$$2 \times 34g \qquad 22.7L \text{ at STP}$$

68 g

On the basis of above equation 22.7 L of O_2 is produced from 68 g H_2O_2 at STP and 10 L of O_2 at

STP is produced from $\frac{68 \times 10}{22.7} g = 29.9g \approx 30g H_2O_2$. Therefore, strength of H_2O_2 in 10 volume H_2O_2

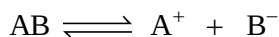
solution = 30 g/L = 3% H_2O_2 solution



25.(6) The Pauli exclusion principle states that no two electrons in an atom can have the same four quantum numbers. In other words, only two electrons may exist in the same atomic orbital, and these electrons must have opposite spins. (a) and (f) violate the Pauli exclusion principle.

Hund's rule states that the most stable arrangement of electrons in subshells is the one with the greatest number of parallel spins. (b), (d) and (e) violate Hund's rule.

26.(-2)

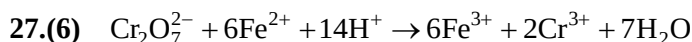


$$1 - 0.075 \quad 0.075 \quad 0.075$$

$$i = 1 + 0.075 = 1.075$$

$$T_f - T_i = iK_b \times m$$

$$T_f = -2^\circ C$$



28.(0) Path A to D is adiabatic.

29.(3) ClF_3 , NH_3 & H_2O are polar.

30.(5) $1s, 2p_x, 4p_z, 2p_y, 4d_{z^2}$.

MATHEMATICS
SECTION-1

$$1.(C) \quad I = \int e^x \left(\frac{1-x^{20}}{(1-x^{10})\sqrt{1-x^{20}}} + \frac{10x^9}{(1-x^{10})\sqrt{1-x^{20}}} \right) dx$$

$$= \int e^x \left(\sqrt{\frac{1+x^{10}}{1-x^{10}}} + \frac{10x^9}{(1-x^{10})\sqrt{1-x^{20}}} \right) dx = \int e^x (f(x) + f'(x)) dx = e^x f(x) + c = e^x \sqrt{\frac{1+x^{10}}{1-x^{10}}} + c$$

$$2.(A) \quad e^{x+y} = y^2$$

$$x + y = 2 \log y$$

$$1 + \frac{dy}{dx} = \frac{2}{y} \frac{dy}{dx}$$

$$\frac{dy}{dx} = \frac{y}{2-y}$$

$$\frac{d^2 y}{dx^2} = \frac{(2-y) \frac{dy}{dx} - y \times \frac{-dy}{dx}}{(2-y)^2} = \frac{2y}{(2-y)^3}$$

$$\left. \frac{d^2 y}{dx^2} \right|_{x=-1} = 2$$

$$3.(A) \quad \sin^2 \beta = \cos \alpha \cdot \sin \alpha \quad \text{and} \quad \cos \alpha < \sin \beta < \sin \alpha$$

$$\Delta = 4 \cot^2 \beta - 4$$

$$= 4 \left(\frac{\cos^2 \beta - \sin^2 \beta}{\sin^2 \beta} \right) = 4 \left(\frac{1 - 2 \sin^2 \beta}{\sin^2 \beta} \right) = 4 \left(\frac{1 - 2 \sin \alpha \cdot \cos \alpha}{\sin^2 \beta} \right) = 4 \left(\frac{\sin \alpha - \cos \alpha}{\sin \beta} \right)^2 > 0$$

$$4.(C) \quad a = n!$$

$$b = \frac{(n-2)!}{(n-12)!}$$

$$c = (n-12)!$$

$$n(n-1) = 182$$

$$= 14 \times 13$$

$$n = 14$$

$$5.(B) \quad P(G) = \frac{1}{4}, P(C) = \frac{1}{8}, P(K) = 1 - \frac{1}{4} - \frac{1}{8} = 1 - \frac{3}{8} = \frac{5}{8}$$

$$P\left(\frac{AC}{G}\right) = \frac{1}{5}, P\left(\frac{AC}{C}\right) = \frac{1}{10}, P\left(\frac{AC}{K}\right) = 1$$

$$P\left(\frac{K}{AC}\right) = \frac{P(K \cap AC)}{P(AC)}$$

$$= \frac{\frac{5}{8} \times 1}{\frac{5}{8} \times 1 + \frac{1}{4} \times \frac{1}{5} + \frac{1}{8} \times \frac{1}{10}} = \frac{\frac{5}{8}}{\frac{5}{8} + \frac{1}{20} + \frac{1}{80}} = \frac{\frac{5}{8}}{\frac{50+4+1}{80}} = \frac{50}{55} = \frac{10}{11}$$

$$6.(B) \quad [\sin x + \cos x] = \left[\sqrt{2} \sin \left(x + \frac{\pi}{4} \right) \right] = \begin{cases} 1 & 0 \leq x \leq \frac{\pi}{2} \\ 0 & \frac{\pi}{2} < x \leq \frac{3\pi}{4} \\ -1 & \frac{3\pi}{4} < x \leq \pi \\ -2 & \pi < x \leq \frac{3\pi}{2} \\ -1 & \frac{3\pi}{2} < x \leq \frac{7\pi}{4} \\ 0 & \frac{7\pi}{4} < x \leq 2\pi \end{cases} \mathbb{K}$$

$$I = 20 \left[\int_0^{\pi/2} 1dx + \int_{\pi/2}^{3\pi/4} 0dx + \int_{3\pi/4}^{\pi} -1dx + \int_{\pi}^{3\pi/2} -2dx + \int_{3\pi/2}^{7\pi/4} -1dx + \int_{7\pi/4}^{2\pi} 0dx \right]$$

$$= 20 \times \left[\frac{\pi}{2} - \pi + \frac{3\pi}{4} - 2 \times \frac{3\pi}{2} + 2\pi - \frac{7\pi}{4} + \frac{3\pi}{2} \right] = 20 \times -\pi = -20\pi$$

7. (A) Using truth table

p	q	r	$p \wedge q$	$\sim(p \wedge q)$	$p \vee r$	$(p \vee r) \vee \sim(p \wedge q)$
T	T	T	T	F	T	T
T	T	F	T	F	T	T
T	F	T	F	T	T	T
T	F	F	F	T	T	T
F	T	T	F	T	T	T
F	T	F	F	T	F	T
F	F	T	F	T	T	T
F	F	F	F	T	F	T

p	$\sim p$	q	$\sim p \wedge q$	$\sim p \vee q$	$(\sim p \wedge q) \vee (\sim p \vee q)$
F	T	T	T	T	T
F	T	F	F	T	T
T	F	T	F	T	T
T	F	F	F	F	F

$p \vee q$	$\sim q$	$((p \vee q) \wedge \sim q)$	$\sim((p \vee q) \wedge \sim q)$
T	F	F	T
F	T	F	T
F	F	F	T
T	T	T	F

$p \rightarrow q$	$q \rightarrow r$	$(p \rightarrow q) \wedge (q \rightarrow r)$
T	T	T
T	F	F
F	T	F
F	F	F

$$\begin{aligned}
 8.(A) \quad f'(x) &= \begin{vmatrix} 1 & 0 & 0 \\ ab & x+b^2 & bc \\ ac & bc & x+c^2 \end{vmatrix} + \begin{vmatrix} x+a^2 & ab & ac \\ 0 & 1 & 0 \\ ac & bc & x+c^2 \end{vmatrix} + \begin{vmatrix} x+a^2 & ab & ac \\ ab & x+b^2 & bc \\ 0 & 0 & 1 \end{vmatrix} \\
 &= (x+b^2)(x+c^2) - b^2c^2 + (x+a^2)(x+c^2) - a^2c^2 + (x+a^2)(x+b^2) - a^2b^2 \\
 &= 3x^2 + 2x(a^2 + b^2 + c^2) < 0 \\
 f(x) &\text{ is decreasing in} \\
 x &\in \left(\frac{-2}{3}(a^2 + b^2 + c^2), 0 \right)
 \end{aligned}$$

$$9.(A) \quad \frac{a}{r} + a + ar = 26 \quad \dots(i)$$

$$\left(\frac{a}{r}\right)^2 + a^2 + a^2r^2 = 364 \quad \dots(ii)$$

$$a^2 \left(\left(r + \frac{1}{r} \right)^2 - 1 \right) = 364$$

$$a \left(r + \frac{1}{r} \right) + a = 26$$

$$\left(r + \frac{1}{r} \right) = \frac{26}{a} - 1$$

$$a^2 \left(\left(\frac{26}{a} - 1 \right)^2 - 1 \right) = 364$$

$$a^2 \left[\left(\frac{26}{a} - 1 + 1 \right) \left(\frac{26}{a} - 1 - 1 \right) \right] = 364$$

$$a^2 \left[\frac{26}{a} \times 2 \left(\frac{13-a}{a} \right) \right] = 364$$

$$13 - a = \frac{364}{52}$$

$$a = 6$$

$$\therefore r = 3$$

$$a_{10} = 2.(3)^9$$

$$10.(A) \quad \int \frac{dx}{x} = \int \frac{ydy}{1+y^2}$$

$$\ell nx = \frac{1}{2} \ell n(1+y^2) + c$$

Passes through (1, 0) so $c = 0$

$$x^2 = 1 + y^2$$

$$x^2 - y^2 = 1 \rightarrow \text{foci} \rightarrow (\pm\sqrt{2}, 0)$$

Tangent at $(\sqrt{2}, 1)$

$$\sqrt{2}x - y = 1$$

Equation of circle touching conic at $(\sqrt{2}, 1)$

$$(x - \sqrt{2})^2 + (y - 1)^2 + \lambda(\sqrt{2}x - y - 1) = 0$$

Passes through $(\sqrt{2}, 0)$ so

$$1 + \lambda = 0 \Rightarrow \lambda = -1$$

Equation of circle

$$(x - \sqrt{2})^2 + (y - 1)^2 - \sqrt{2}x + y + 1 = 0$$

$$x^2 + y^2 - 3\sqrt{2}x - y + 4 = 0$$

$$11.(A) \lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} \frac{e^{0+h} - 3}{0 + x + 1} = -2$$

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{h \rightarrow 0} \frac{e^{-1-h} - 3}{-1 - h + 1} = \lim_{h \rightarrow 0} \frac{e^{-1-h} - 3}{-h} = +\infty$$

12.(B) Equation of normal to ellipse

$$\frac{ax}{\cos \theta} - \frac{by}{\sin \theta} = a^2 - b^2$$

For point A

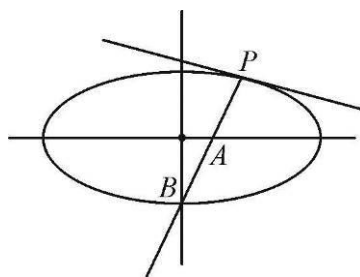
$$y = 0$$

$$x = \cos \theta \left(\frac{a^2 - b^2}{a} \right)$$

For point B

$$x = 0$$

$$y = \sin \theta \left(\frac{b^2 - a^2}{b} \right)$$



$$(a \cos \theta, b \sin \theta) \left(\cos \theta \left(\frac{a^2 - b^2}{a} \right), 0 \right) \left(0, \sin \theta \left(\frac{b^2 - a^2}{b} \right) \right)$$

$$0 = \frac{b \sin \theta \times 1 + 3 \sin \theta \left(\frac{b^2 - a^2}{b} \right)}{4}$$

$$b^2 + 3(b^2 - a^2) = 0$$

$$4b^2 - 3a^2 = 0 \rightarrow \frac{b^2}{a^2} = \frac{3}{4} \rightarrow \text{can be directly used}$$

$$e = \sqrt{1 - \frac{3}{4}} = \frac{1}{2}$$

$$\begin{aligned}
 13.(A) \quad \lim_{x \rightarrow \frac{\pi}{2}^-} f(x) &= \lim_{h \rightarrow 0} \left(1 + \cos^{-1} \left\{ \cot \left(\frac{\pi}{2} - h \right) \right\} \right) \\
 &= \lim_{h \rightarrow 0} 1 + \cos^{-1} \{ \tanh \} = \lim_{h \rightarrow 0} 1 + \cos^{-1}(h) = 1 + \frac{\pi}{2} \\
 \lim_{x \rightarrow \frac{\pi}{2}^+} f(x) &= \lim_{h \rightarrow 0} \pi \left[\frac{\pi}{2} + h \right] + 1 = \pi + 1 \\
 \text{Jump} &= \pi + 1 - 1 - \frac{\pi}{2} = \frac{\pi}{2}
 \end{aligned}$$

$$\begin{aligned}
 14.(A) \quad \frac{1}{m} \sum (y_i + 3)^2 &= 16 \quad \dots(i) \\
 \frac{1}{m} \sum (y_i - 3)^2 &= 8 \quad \dots(ii)
 \end{aligned}$$

Adding (i) and (ii)

$$\frac{1}{m} \sum_{i=1}^m (2y_i^2 + 18) = 24$$

$$\frac{2}{m} \sum_{i=1}^m y_i^2 + 18 = 24 \quad ; \quad \frac{\sum_{i=1}^m y_i^2}{m} = 3$$

Subtracting (i) and (ii)

$$\frac{12}{m} \sum_{i=1}^m y_i = 8$$

$$\frac{\sum_{i=1}^m y_i}{m} = \frac{8}{12} \quad ; \quad \sigma = \sqrt{\frac{\sum y_i^2}{m} - \left(\frac{\sum y_i}{m} \right)^2} = \sqrt{3 - \left(\frac{2}{3} \right)^2} = \sqrt{\frac{23}{9}} = \frac{\sqrt{23}}{3}$$

15.(B) If d.r.'s of line of intersection are (a, b, c) then

$$2a - 3b + 9c = 0 \quad \dots(i)$$

$$4a - 2b + 7c = 0 \quad \dots(ii)$$

$$\frac{a}{-21+18} = \frac{b}{36-14} = \frac{c}{-4+12} \quad ; \quad \frac{a}{-3} = \frac{b}{22} = \frac{c}{8}$$

$$\text{Equation of line } \frac{x}{-3} = \frac{y-1}{22} = \frac{z+1}{8}$$

$$16.(A) \quad a = A(R)^{p-1} \rightarrow \log a = \log A + (p-1) \log R$$

$$b = A(R)^{q-1} \rightarrow \log b = \log A + (q-1) \log R$$

$$c = A(R)^{r-1} \rightarrow \log c = \log A + (r-1) \log R$$

Angle between two vectors

$$\vec{\alpha} \cdot \vec{\beta} = 3 \sum (q-r) \log a$$

$$= 0$$

$$\text{So, angle between vectors} = \frac{\pi}{2}$$

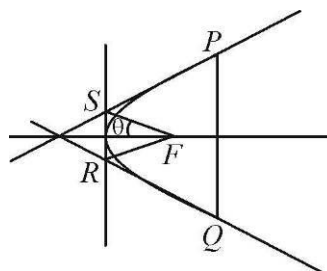
17.(C) $P(at_1^2, 2at_1)$ $t_2 = -t_1$

$Q(at_2^2, 2at_2)$

Equation of tangent at P $t_1y = x + at_1^2$

$S \equiv (0, at_1)$

$R \equiv (0, -at_1)$



Area of trapezium PQRS

$$= \frac{1}{2}(2at_1 + 4at_1) \times at_1^2 = 24a^2$$

$$= 3a^2t_1^3 = 24a^2$$

$$t_1^3 = \frac{24}{3} = 8$$

$$t_1 = 2$$

$$\tan \theta = \frac{at_1}{a} = t_1 = 2$$

Angle subtended by SR at focus

$$\tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta} = \frac{4}{1 - 4} = \frac{-4}{3} ; \alpha = \tan^{-1}\left(\frac{-4}{3}\right)$$

18.(C) $B + C = 180^\circ - A$

$B - C = 60^\circ$

$B = \frac{2\pi}{3} - \frac{A}{2}$

$C = \frac{\pi}{3} - \frac{A}{2}$

Area of triangle ABC $= \frac{1}{2}bc \sin A$

Using sine rule

$$b = \frac{a \sin B}{\sin A}, c = \frac{a \sin C}{\sin A}$$

$$12 = \frac{1}{2} \times \frac{a \sin B}{\sin A} \times \frac{a \sin C}{\sin A} \times \sin A$$

$$24 = a^2 \frac{\sin B \cdot \sin C}{\sin A}$$

$$\sin A = \frac{16}{24} \times \sin B \sin C = \frac{2 \sin B \cdot \sin C}{3} = \frac{\cos(B - C) - \cos(B + C)}{3}$$

$$\sin A = \frac{\cos \frac{\pi}{3} - \cos(\pi - A)}{3}$$

$$3 \sin A = \frac{1}{2} + \cos A ; \quad 3 \sin A - \frac{1}{2} = \cos A$$

$$9 \sin^2 A + \frac{1}{4} - 3 \sin A = 1 - \sin^2 A$$

$$10 \sin^2 A - 3 \sin A - \frac{3}{4} = 0 ; \sin A = \frac{3 \pm \sqrt{9 + 30}}{20} = \frac{3 + \sqrt{39}}{20}, \frac{3 - \sqrt{39}}{20}$$

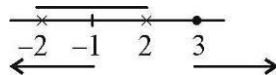
$A < 30^\circ$

19.(C) $P \Rightarrow x \in (-2, 2)$

$Q \Rightarrow x - 1 \geq 2 \text{ or } x - 1 \leq -2$

$x \geq 3 \text{ or } x \leq -1$

$P \cup Q = R - [2, 3)$



20.(B) equation of plane $\begin{vmatrix} x-2 & y-1 & z-1 \\ 0 & 2 & 2 \\ 1 & 1 & 3 \end{vmatrix} = 0$

$2x + y - z - 4 = 0$

Point on the line $\frac{x-1}{1} = \frac{y-2}{2} = \frac{z-3}{3} = r$ is $(1, 2, 3)$

Reflection of $(1, 2, 3)$ in the plane

$\frac{x_1-1}{2} = \frac{y_1-2}{1} = \frac{z_1-3}{-1} = \frac{-2(2+2-3-4)}{(2)^2 + (1)^2 + (-1)^2}$

$\frac{x_1-1}{2} = \frac{y_1-2}{1} = \frac{z_1-3}{-1} = \frac{6}{6}$

$x_1 = 3$

$y_1 = 3$ (3, 3, 2)

$z_1 = 2$

Point of intersection of line and plane

$2(1+r) + 1(2+2r) - (3+3r) - 4 = 0$

$r = 3$

Point of intersection $(4, 8, 12)$

Equation of line of reflection

$\frac{x-3}{4-3} = \frac{y-3}{8-3} = \frac{z-2}{12-2} ; \frac{x-3}{1} = \frac{y-3}{5} = \frac{z-2}{10}$

SECTION-2

21.(0) $2^{2\log_5 4 \cdot \log_4 x + x \log_2 5} + 5^x + x^5 \cdot x^{\log_5 4} + x^5 = 0$

$2^{2\log_5 x} \times 2^{x \log_2 5} + 5^x + x^5 (4^{\log_5 x} + 1) = 0$

$5^x (2^{\log_5 x})^2 + 5^x + x^5 ((2^{\log_5 x})^2 + 1) = 0$

$(5^x + x^5) \left(\underbrace{(2^{\log_5 x})^2 + 1}_{\text{Always } > 0} \right) = 0$

$5^x + x^5 = 0$ Only for $x < 0$

Only solution can be for $x < 0$

But for log to define $x > 0$

So, no solution exists

22.(2) $z^2 - (3+i)z + (m+2i) = 0$

Let one real root is k

Then $k^2 - (3+i)k + (m+2i) = 0$

Real part = 0 imaginary part = 0

$k^2 - 3k + m = 0$ $-k + 2 = 0$

$m = 2$ $k = 2$

Let non real root is z_1

$2 + z_1 = 3 + i$

$z_1 = 3 + i - 2$

$z_1 = 1 + i$

Product = $2 \times 1 = 2$

23.(1.50)
$$\lim_{n \rightarrow \infty} n^{-n^2} \left((n+1) \left(n + \frac{1}{3} \right) \left(n + \frac{1}{3^2} \right) \dots \left(n + \frac{1}{3^{n-1}} \right) \right)^n$$

$$= \lim_{n \rightarrow \infty} \left(\frac{n+1}{n} \right)^n \cdot \left(\frac{n + \frac{1}{3}}{n} \right)^n \dots \left(\frac{n + \frac{1}{3^{n-1}}}{n} \right)^n = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \right)^n \left(1 + \frac{1}{3n} \right)^{\frac{3n}{3}} \dots \left(1 + \frac{1}{3^{n-1} \cdot n} \right)^{\frac{3^{n-1} \cdot n}{3^{n-1}}}$$

$$= e^{\frac{1}{3}} \cdot e^{\frac{1}{9}} \dots e^{\frac{1}{3^{n-1}}} = e^{1 + \frac{1}{3} + \frac{1}{3^2} + \dots + \frac{1}{3^{n-1}}} = e^{\frac{1}{1 - \frac{1}{3}}} = e^{\frac{3}{2}} ; \quad k = 1.50$$

24.(30) $C_1 \equiv (1,1), C_2 \equiv (9,6), r_1 = 1, r_2 = 2$

$C_1 M_1 \geq r_1$

$C_2 M_2 \geq r_2$

$\left| \frac{3+4-k}{5} \right| \geq 1$

$|7-k| \geq 5$

$k-7 \geq 5$

$k \geq 12 \quad \dots(i)$

$|27+24-k| \geq 10$

$51-k \geq 10$

$k \leq 41 \quad \dots(ii)$

From (i) and (ii)

$k \in [12, 41]$

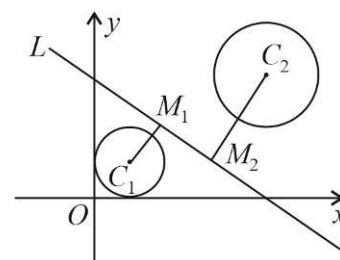
C_1 is below the line $3x + 4y - k = 0$

$7 - k < 0$

C_2 lies above the line $3x + 4y - k = 0$

$51 - k > 0$

Number of integral values = 30



25.(4) $X = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \rightarrow X^2 = \begin{bmatrix} a^2 + bc & ab + bd \\ ac + cd & bc + d^2 \end{bmatrix}$

$a^2 + bc = 1 \quad \dots(i)$

$ab + bd = 1 \Rightarrow b(a + d) = 1$

$ac + cd = 2 \quad c(a + d) = 2$

$\frac{b}{c} = \frac{1}{2} \rightarrow c = 2b$

$$bc + d^2 = 3 \quad \dots(ii)$$

$$(ii) - (i)$$

$$d^2 - a^2 = 2$$

$$(d - a)(d + a) = 2 ; \quad (d - a) = 2b$$

$$a + d = \frac{1}{b} ; \quad d = b + \frac{1}{2b} ; \quad a = \frac{1}{2b} - b$$

$$a^2 + bc = 1$$

$$b^2 + \frac{1}{4b^2} - 1 + 2b^2 = 1 ;$$

$$3b^2 + \frac{1}{4b^2} - 2 = 0$$

$$b^2 = x$$

$$3x + \frac{1}{4x} - 2 = 0 \rightarrow 12x^2 - 8x + 1 = 0$$

$$b^2 = \frac{1}{6}, b^2 = \frac{1}{2} \quad \text{Four such matrices exist}$$

$$26.(1) \cos(2 \tan^{-1}(x+1)) = \frac{x}{2}$$

$$\cos 2\theta = \frac{1 - \tan^2 \theta}{1 + \tan^2 \theta} ; \quad \frac{1 - (x+1)^2}{1 + (x+1)^2} = \frac{x}{2}$$

$$2(1+x+1)(1-x-1) = x(x^2 + 2x + 2) \Rightarrow 2(x+2)(-x) = x(x^2 + 2x + 2)$$

$$x(x^2 + 2x + 2 + 2x + 4) = 0$$

$$x(x^2 + 4x + 6) = 0$$

$$x(x^2 + 4x + 6) = 0$$

$$x = 0 \text{ is the only solution}$$

$$27.(15) (1+x-x^2)^5 = (1+x(1-x))^5$$

$$= {}^5C_0 + {}^5C_1 x(1-x) + {}^5C_2 x^2(1-x)^2 + {}^5C_3 x^3(1-x)^3 + {}^5C_4 x^4(1-x)^4 + {}^5C_5 x^5(1-x)^5$$

$$\text{Coefficient of } x^4 = {}^5C_2 + {}^5C_3(-3) + {}^5C_4 = 10 - 30 + 5 = -15$$

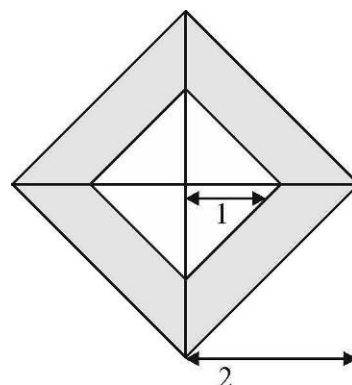
$$28.(16) 1 \leq \left| \frac{x+y}{\sqrt{2}} \right| + \left| \frac{x-y}{\sqrt{2}} \right| \leq 3$$

$$1 \leq |X| + |Y| \leq 3 \quad (\text{Rotation through an angle of } \frac{\pi}{4})$$

$$\text{Area} = 4 \left(\frac{1}{2} \times 3 \times 3 - \frac{1}{2} \times 1 \times 1 \right)$$

$$= 4 \left[\frac{9}{2} - \frac{1}{2} \right]$$

$$= 16(\text{unit})^2$$



$$29.(5) \quad f(x) = \int_1^x \left(t + \frac{1}{t} \right) dt$$

$$g(x) = x + \frac{1}{x} \text{ for } x \in \left[\frac{1}{2}, 3 \right]$$

$$g\left(\frac{1}{2}\right) = \frac{5}{2}, \quad g(3) = \frac{10}{3}$$

$$\text{Let } P(\alpha, g(\alpha)), \quad \alpha \in \left[\frac{1}{2}, 3 \right]$$

By L.M.V.T.

$$g'(\alpha) = \frac{g(3) - g\left(\frac{1}{2}\right)}{3 - \frac{1}{2}}$$

$$1 - \frac{1}{\alpha^2} = \frac{\frac{10}{3} - \frac{5}{2}}{\frac{5}{2}} = \frac{4-3}{3} = \frac{+1}{3}$$

$$\alpha^2 = \frac{3}{2} \Rightarrow \alpha = \sqrt{\frac{3}{2}}$$

$$g(\alpha) = \sqrt{\frac{3}{2}} + \frac{1}{\sqrt{3/2}}$$

$$\lambda = \frac{5}{\sqrt{6}}$$

$$\lambda\sqrt{6} = 5$$

30.(0.50)

$$\Delta = abc \begin{vmatrix} \frac{-bc}{a} & b+a & c+b \\ a+c & \frac{-ac}{b} & c+a \\ a+b & a+b & \frac{-ab}{c} \end{vmatrix} ; \quad \Delta = \frac{abc}{abc} \begin{vmatrix} -bc & ab+ac & ac+ab \\ ab+bc & -ac & bc+ab \\ ac+bc & ac+bc & -ab \end{vmatrix}$$

$$\Delta = \begin{vmatrix} -bc & ab+ac & ac+ab \\ ab+bc & -ac & ab+bc \\ ac+bc & ac+bc & -ab \end{vmatrix} ; \quad \Delta = \begin{vmatrix} -bc-ab-ac & 0 & ac+ab \\ ab+bc+ac & -ab-bc-ac & ab+bc \\ 0 & ab+bc+ac & -ab \end{vmatrix} \begin{matrix} C_1 \rightarrow C_1 - C_2 \\ C_2 \rightarrow C_2 - C_3 \end{matrix}$$

$$= (ab+bc+ac)^2 \begin{vmatrix} -1 & 0 & ac+ab \\ 1 & -1 & ab+bc \\ 0 & 1 & -ab \end{vmatrix} R_1 \rightarrow R_1 + R_2 + R_3$$

$$\Delta = (ab+bc+ac)^3 = 27$$

$$ab+bc+ac=3 ; \quad a^2+c^2+b^2=2$$

$$a+b+c = \frac{-q}{p} ; \quad (a+b+c)^2 = \frac{q^2}{p^2} ; \quad 2+2 \times 3 = \frac{q^2}{p^2} \Rightarrow \frac{q^2}{p^2} = 8 ; \quad \frac{p}{q} = \frac{-1}{2\sqrt{2}} ; \quad \frac{\sqrt{2}p}{q} + 1 = \frac{1}{2}$$